

Vignesh Chowdary Pamulapati

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SUMMARY

Software Engineer with 2+ years building and optimizing production systems across enterprise healthcare and SaaS environments. Improved reliability and performance at Optum on distributed claims workflows processing 2M+ daily healthcare transactions; previously engineered multi-tenant APIs serving 250K+ SMB customers across 50+ countries at Nowfloats. Experienced in backend systems, distributed workflows, production reliability, and performance engineering.

EDUCATION

Northern Arizona University Aug 2024 – Dec 2025
M.S. in Computer Science | GPA: 3.82 / 4.0 Flagstaff, AZ
Relevant Coursework: Software Engineering, Distributed Systems, Cloud Computing, Advanced Intelligent Systems, Data Structures & Algorithms

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, C/C++
Frontend: React.js, Next.js, Tailwind CSS, HTML5
Backend & APIs: FastAPI, Node.js, REST APIs, WebSocket, Async/Await, JWT, RBAC
Databases & Caching: PostgreSQL, MySQL, MongoDB, SQLite, Redis
DevOps & Cloud: Docker, AWS (EC2, Lambda, S3), GitHub Actions, CI/CD, Git, Linux
Practices: System & API Design, Distributed Systems, Agile/Scrum, Unit/Integration Testing, Code Reviews

EXPERIENCE

Optum Jan 2026 – Present
Software Engineer, Enterprise Systems San Jose, CA (Hybrid)

- Redesigned retry handling and isolated blocking service dependencies in claims adjudication workflows processing 2M+ daily transactions; cut **claim processing failures by 30%** and restored same-day completion reliability across peak submission windows.
- Optimized nightly healthcare data pipelines processing 6M+ records through execution-plan tuning, query optimization, and workload parallelization; restored downstream reporting SLA compliance and unblocked dependent data teams from daily delays.
- Automated failed claims reconciliation through fault-tolerant retry and transaction validation workflows integrated with internal REST APIs; eliminated **40% of manual remediation work** and shortened average claim recovery time from days to under 12 hours.
- Diagnosed recurring production failures across distributed provider and claims services by tracing timeout patterns and Oracle transaction states; identified cascading dependency failures that reduced recurring incidents and improved system reliability.

Optum May 2025 – Dec 2025
Software Engineering Intern, Enterprise Systems Flagstaff, AZ (Remote)

- Improved provider eligibility **API response time by 25%** by profiling **SQL** bottlenecks and streamlining downstream service calls.
- Built internal monitoring workflows consolidating failed healthcare transactions and service exceptions into centralized operational views; cut average troubleshooting time from **45 to under 20 minutes**, measured across recurring incidents before and after deployment.
- Automated deployment readiness checks across lower environments by validating API availability, configuration mismatches, and database connectivity; saved 6 engineering hours per sprint and improved release reliability.

Nowfloats Technologies Jan 2023 – Jul 2024
Software Engineer, Backend Platform Hyderabad, India

- Engineered REST APIs for merchant onboarding, analytics, and profile management on a multi-tenant SaaS platform serving 250K+ SMB customers across 50+ countries; optimized PostgreSQL and MongoDB through compound indexing, cutting **API latency by 40%** (800ms → 480ms) with zero downtime.
- Rebuilt merchant-facing React dashboards on a platform connecting **6M+** monthly consumers; applied memoization and eliminated redundant API calls, **cutting page load by 25%** and resolving 5 user-reported defects.
- Designed JWT authentication and RBAC across admin, staff, and viewer roles for a multi-tenant merchant platform; enforced permission isolation and eliminated cross-tenant data exposure in a live production system.
- Strengthened backend reliability across payments, search, and reporting services by expanding automated unit and integration test coverage for critical **API workflows**; reduced regression defects and improved release stability for production deployments.

PROJECTS

DocMind: Retrieval Agent with Evaluation Framework github.com/AI-Knowledge-Base
Python, FastAPI, LangChain, Hugging Face, Groq API, MongoDB Vector Search, PostgreSQL, Next.js, AWS, Docker

- Built a retrieval evaluation harness that scores top-k precision across chunk sizes and overlap windows against a **MongoDB Vector Search** index, with document metadata and session history persisted in PostgreSQL; switching from fixed 500-token chunking to variable 350–450-token windows with 10–15% overlap pushed **precision from 60% to 85%**.
- Architected a citation-grounded generation pipeline: **FastAPI backend** injects source document references into every Groq API call via **LangChain**, ensuring every answer is traceable to a specific document chunk and preventing unsupported generation.

LinguaLink: Real-Time Multilingual Platform github.com/LinguaLink
TypeScript, Node.js, Socket.IO, Hugging Face, MongoDB, Firebase, React, Tailwind CSS

- Built a real-time translation pipeline: Node.js/Socket.IO backend routes messages through Hugging Face NLP for 15+ languages; response caching keyed by language pair **reduced inference calls by over 60%** for high-frequency language pairs under concurrent load.
- Implemented RBAC workspaces with group and direct messaging over live **WebSocket** connections; JWT auth, MongoDB schema design, and **Firebase deployment** with isolated production and staging environments.